FARM MANAGEMENT SYSTEM

JAWAHAR SCIENCE COLLEGE, NEYVELI

DERPARTMENT OF COMPUTER SCINCE AND APPLICATION

COURSE: BACHELORE OF COMPUTER SCIENCE

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ABSTRACT

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The **Farm Management System** (FMS) is a Salesforce-based application developed to digitize and manage agricultural operations efficiently. The system provides a centralized platform to store and manage data related to farms, crops, farmers, and buyers.

This application helps farmers and farm managers maintain accurate records, track farming activities, and monitor sales and production. By using Salesforce features such as custom objects, validation rules, relationships, reports, and dashboards, the system ensures organized data management and real-time insights.

The Farm Management System reduces manual paperwork, improves data accuracy, and supports better decision-making through analytics and reports. The project aims to simplify farm management processes and enhance productivity by integrating modern cloud technology into agricultural management.

INTRODUCTION

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Agriculture plays a vital role in the economy and food production of many countries. Managing farming activities efficiently is essential for improving productivity and maintaining proper records of crops, farms, farmers, and buyers. Traditional farming management methods mainly depend on manual record-keeping, which can be time-consuming and prone to errors.

The **Farm Management System (FMS)** is a cloud-based application developed using the Salesforce platform to digitize and simplify agricultural management. The system helps store, manage, and analyze farm-related data in an organized way. It allows users to maintain information about farms, crops, farmers, and buyers while providing insights through reports and dashboards.

By using Salesforce features such as custom objects, validation rules, relationships, and analytics tools, the Farm Management System improves efficiency, reduces manual work, and supports better decision-making. The system aims to modernize farm management by integrating technology into agricultural operations.

Project view:

The Farm Management System reduces manual record-keeping, improves data accuracy, and enhances decision-making capabilities. It supports farmers and farm managers in tracking operations efficiently, maintaining transparency, and improving overall agricultural management.

Purpose:

The main purpose of this project is to:

- Digitize and automated agricultural operation
- Track agricultural activities efficiently.
- Provide real-time insights through reports and dashboards.
- Monitor sales, Expenses, and financial transactions.
- Enhance overall productivity and operational efficiency.

REQUIREMENTS ANALYSIS PHASE

Customer Journey Map

Stages:

1.Login

2.Access Farm Management App

3.Enter Farm and Crop data

4.Resigter Farmers and buyer

5.Generate Reports

6.Analyze Results

7.Dicision Making

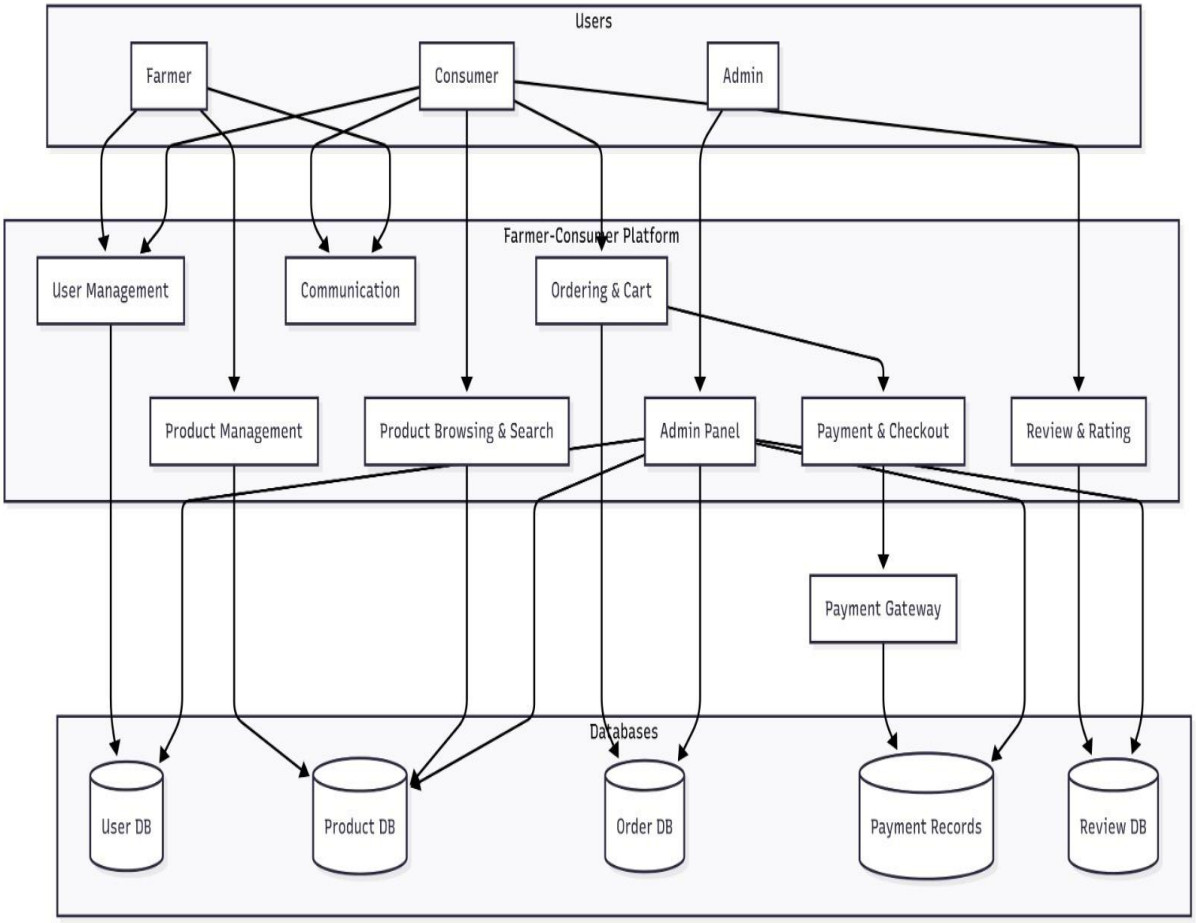
DATA FLOW DIAGRAM

Data flow:

The data flow of the Farm Management System (FMS) follows a structured process within the Salesforce platform:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards

ARCHITECTURE DIAGRAM



PROJECT PLANNIGN

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

PROJECT DEVELOPMENT PHASE

Project description

Farm Management System (FMS) is developed using the Salesforce platform to provide a comprehensive solution for managing agricultural operations digitally. The system is designed to centralize and organize farm-related data such as farms, crops, farmers, buyers, and sales information.

MILESTONE 1-SALESFORCE

Sales force:

Salesforce is a cloud-based **Customer Relationship Management (CRM)** platform that enables organizations to manage data, automate processes, and generate insights. It provides tools for application development, automation, reporting, and secure cloud storage.

Activity 1:

Creating a Salesforce Developer Org :

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search Developer.salesforce.com/signup

2. Enter the following details like First name, last name, Email, Role, Company,

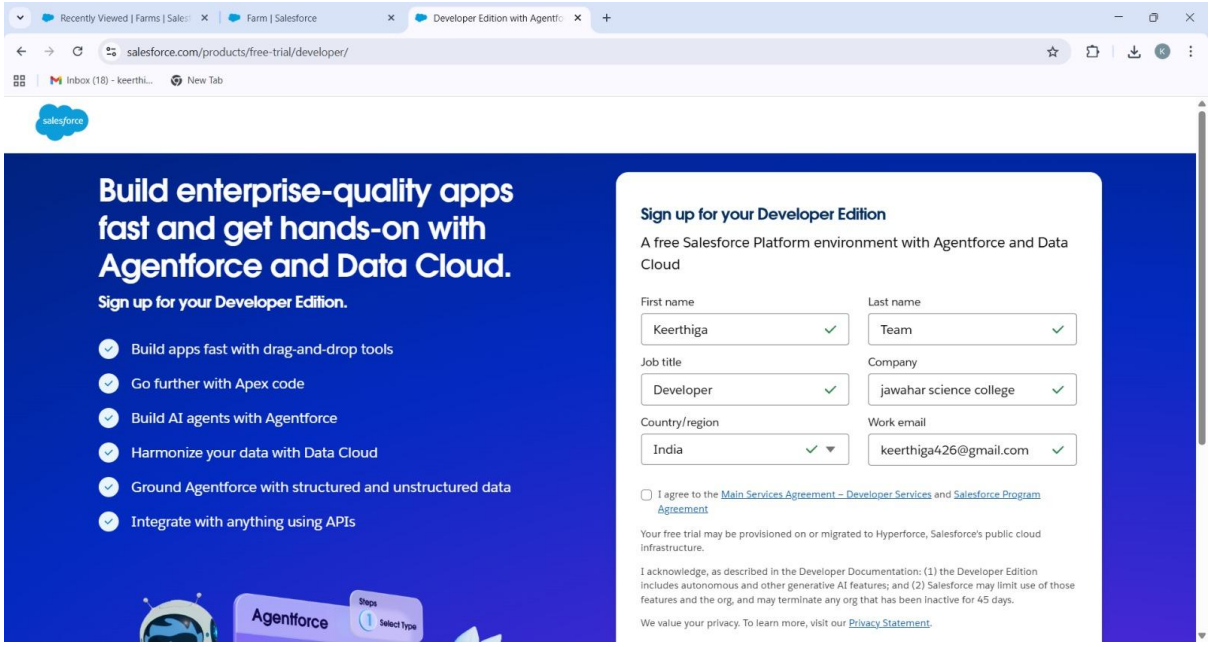
Country/Region, Postal code, and Username must be unique.

3. Click sign me up, after a few min you will reserve a mail salesforce org and by using

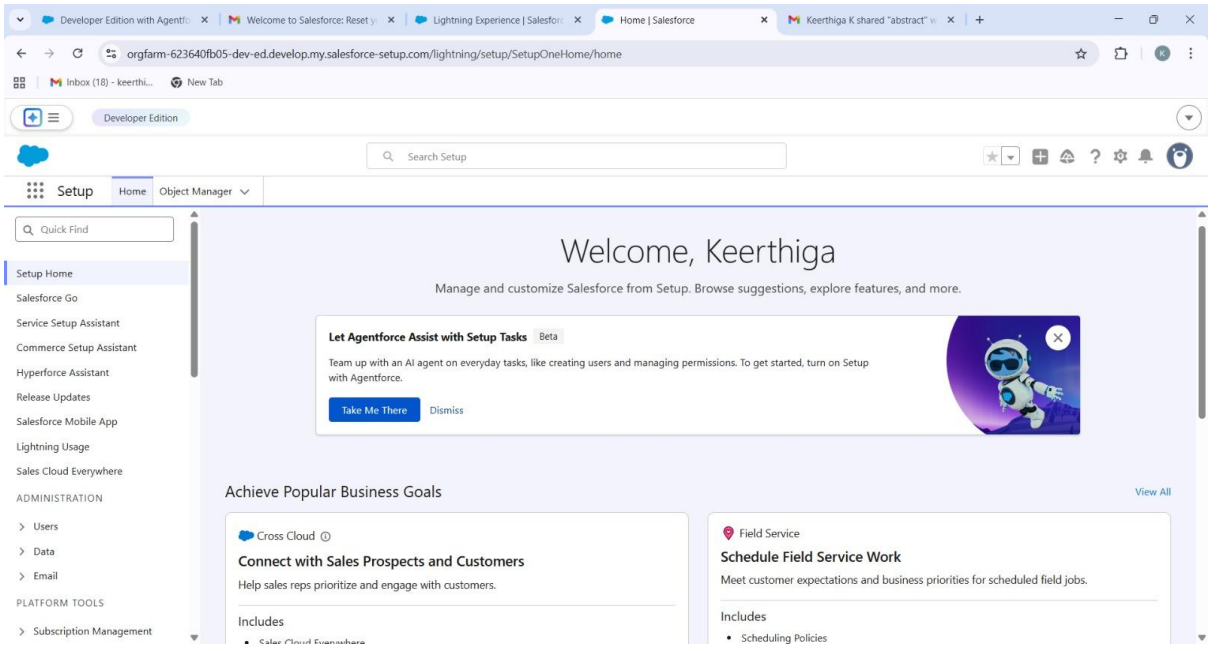
the verify account link you can create your new password.

4. Click save.

5. Search login.salesforce.com



By using username and password you can into the Salesforce org. The homepage will appear below



MILESTONE 2-OBJECT:

Object

An Object in Salesforce is a database table used to store data in an organized manner. It consists of records (rows) and fields (columns), where each record represents a specific entry and each field stores particular information about that entry.

In This Application We Use 4 Custom Objects:

1.Farm

2.Crop

3.Farmer

4.Buyer

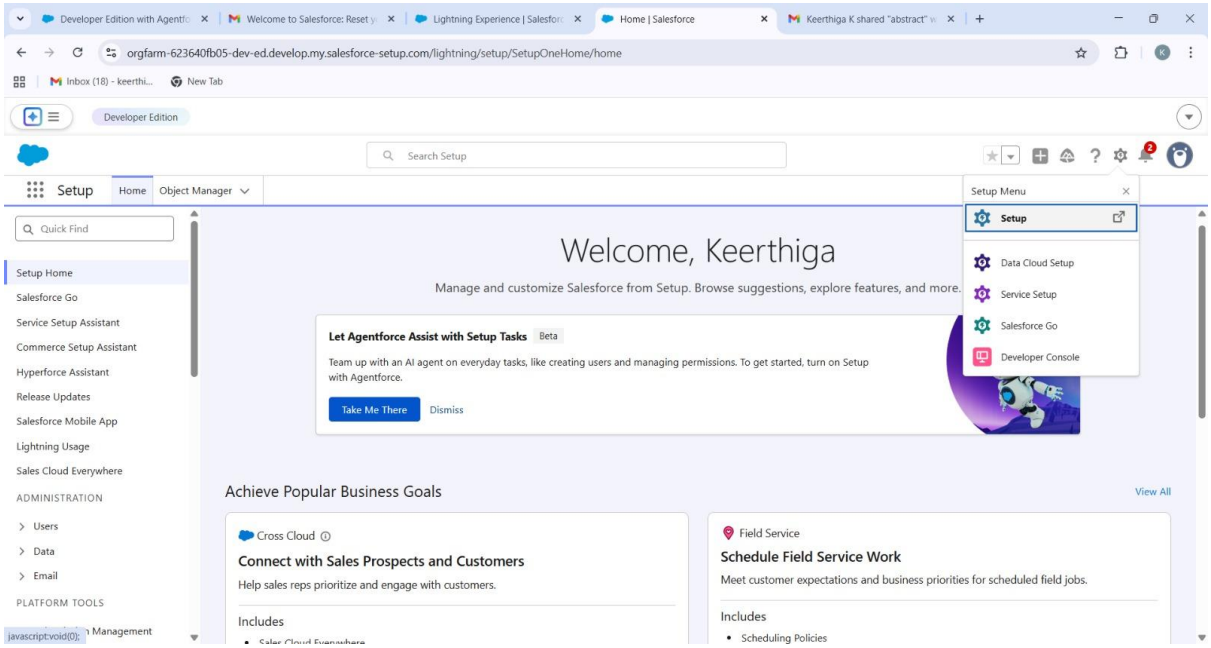
Activity 1:

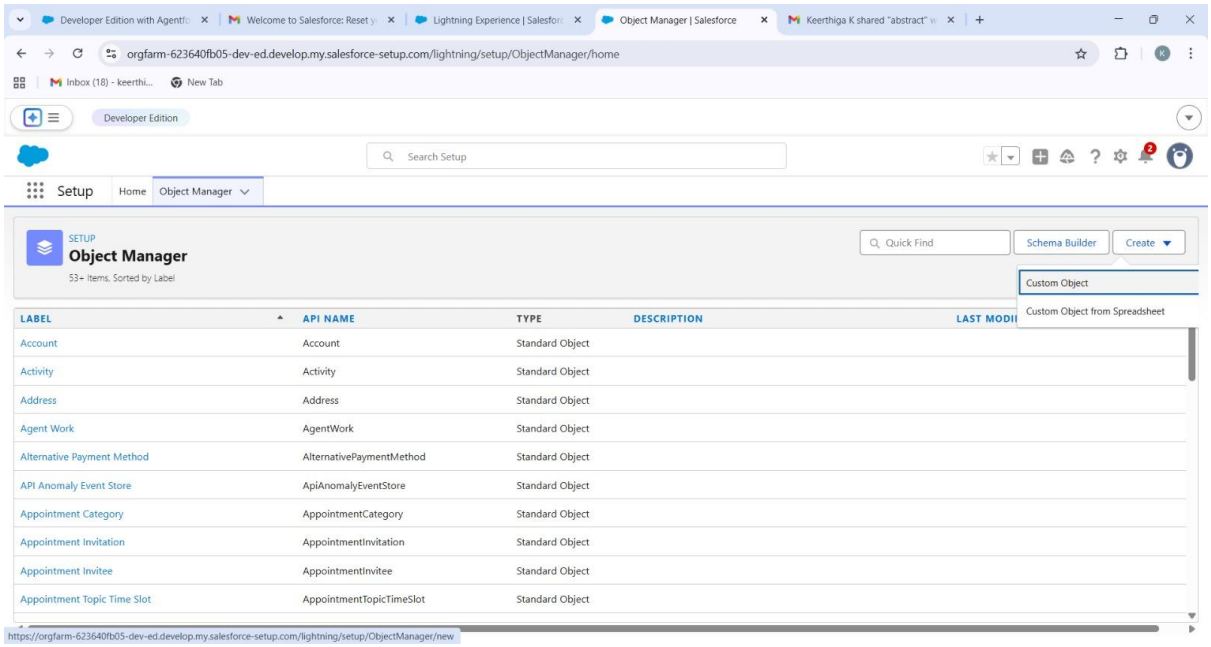
Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.

2. Click create, select custom object.



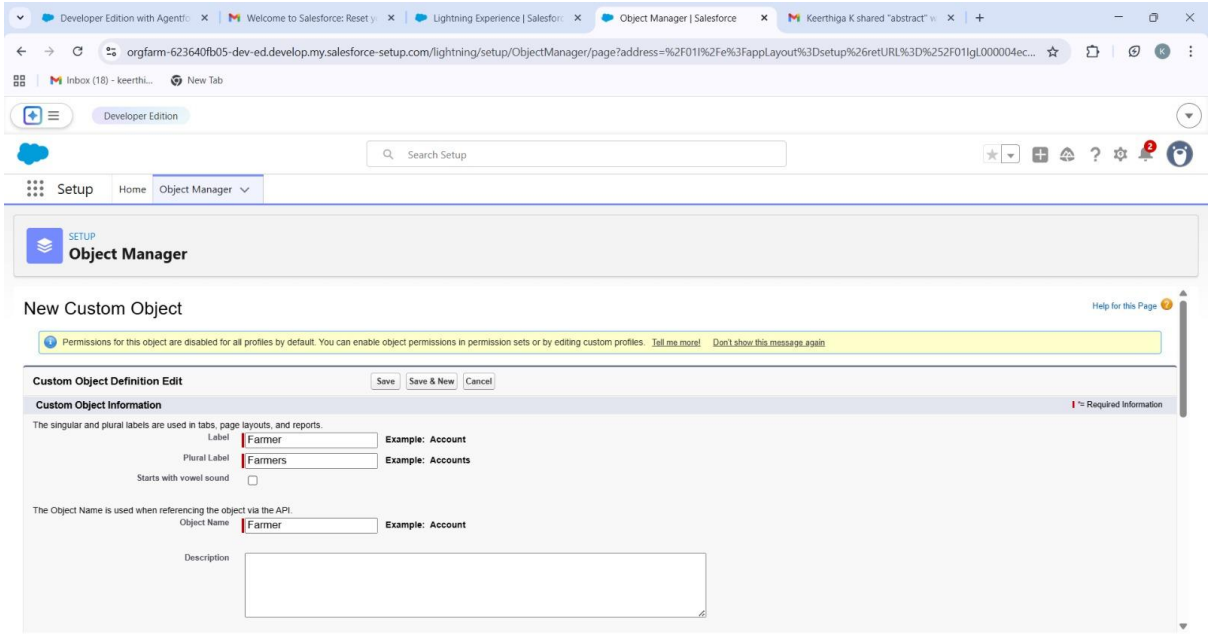


3. Fill in the label as " Farm ".

4. Fill in the plural label as " Farms".

5. Record name: Farm Name"

6. Select the data type as "Text".



7. In the Optional Features section, select Allow Reports and Track Field History.

8. In the Deployment Status section, ensure Deployed is selected.

9. In the Search Status section, select Allow Search.

10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.

11. Leave everything else as is and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Drop down click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:

b. Label: Crop

c. Plural Label: Crops

d. Record Name: Crop Name

e. Select the data type as "Text".

f. Check the Allow Reports checkbox g. Check the Allow Search checkbox

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout if. Click, Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Drop down click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:
b. Label: Farmer

c. Plural Label: Farmers

d. Record Name: Farmer Name e. Select the data type as "Text".

f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.

2. Click on the object manager tab just beside the home tab.

3. After the above steps, have a look on the extreme right you will find a Create Drop down click on that and select Custom Object.

a. On the Custom Object Definition page, create the object as follows:

b. Label: Buyer

c. Plural Label: Buyers

d. Record Name: Buyer Name

e. Select the data type as "Text".

f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout, Click Save.

MILESTONE-3

Tab: Tab

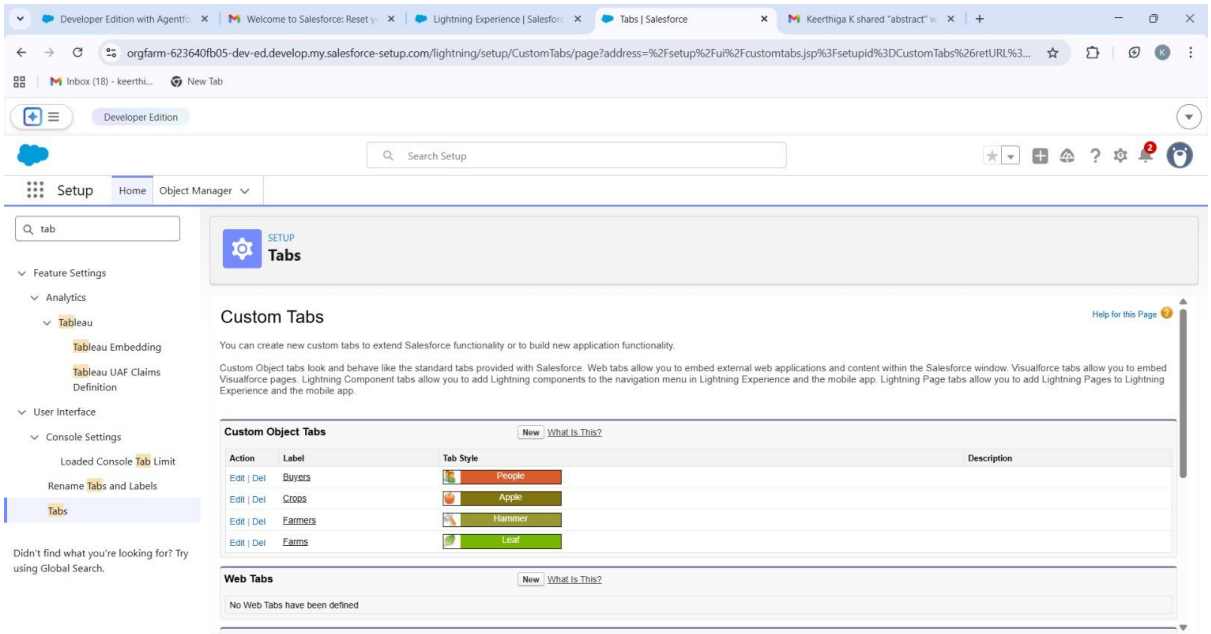
Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

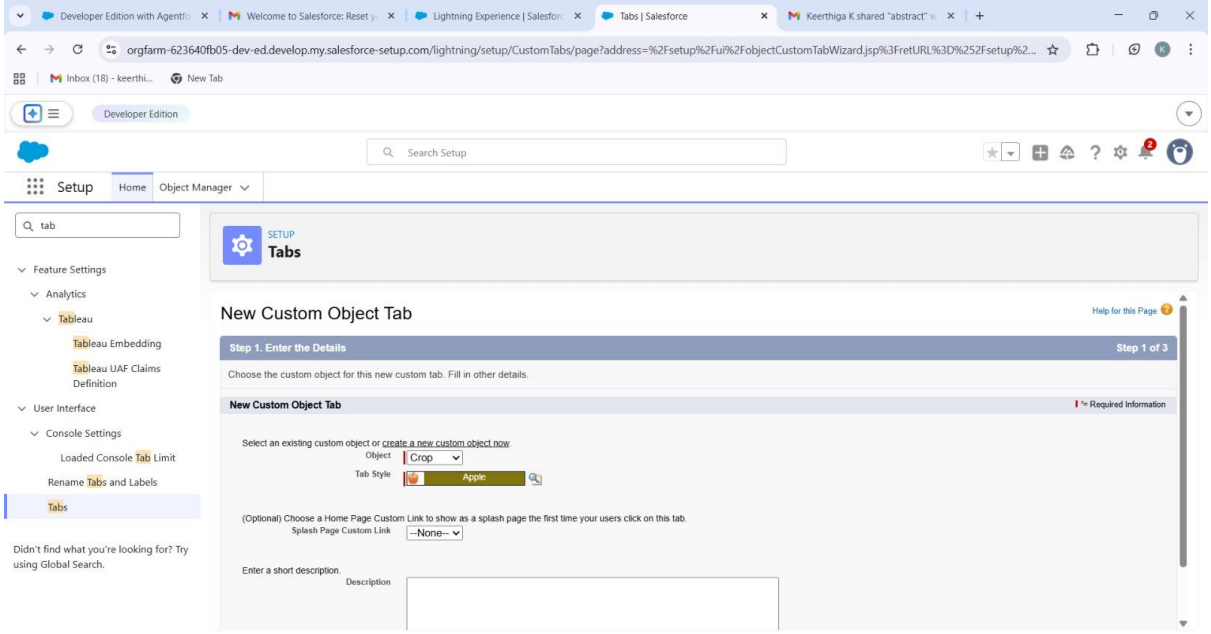
Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.



2.Under custom object tabs, click New.



2.For Object, select Farm.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Crop.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Farmer.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Buyer

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

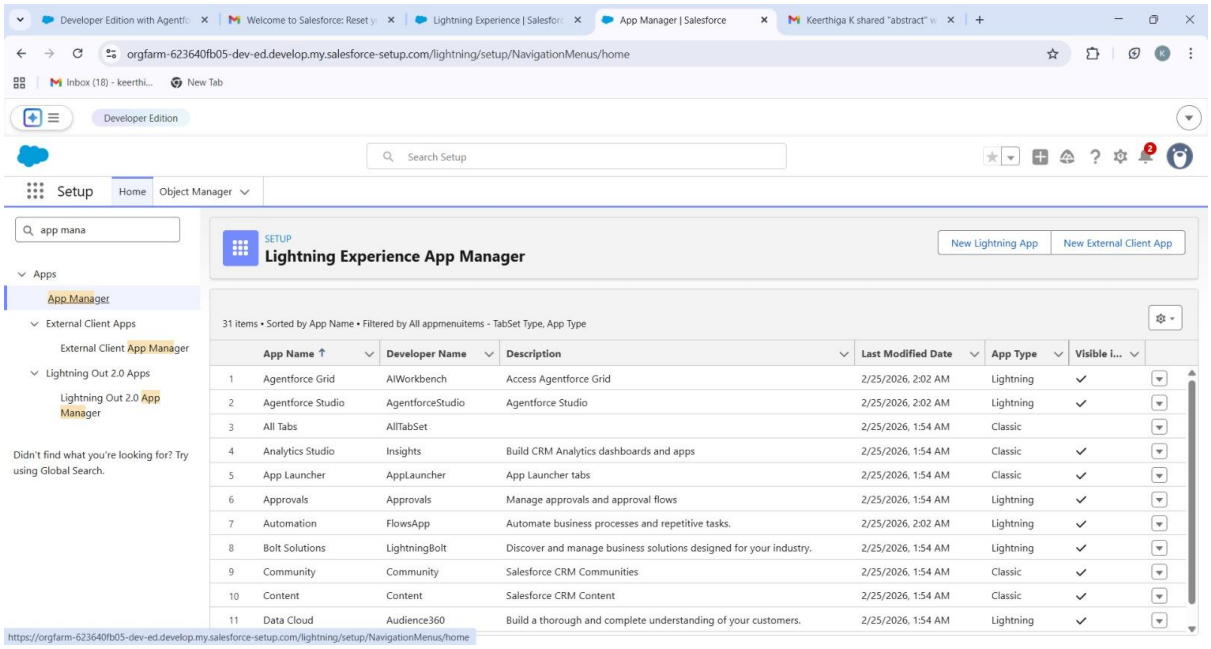
MILESTONE-4/ Lighting app

App:

An App in Salesforce is a collection of related components such as objects, tabs, reports, dashboards, and other features grouped together to perform a specific business function.

Activity 1:

Create the Farm Management system app



● From Setup, enter App Manager in the Quick Find and select App Manager.

● Click New Lightning App. Farm Management System as the App Name, then click Next

● Under App Options, leave the default selections and click Next.

● Under Utility Items, leave as is and click Next.

● From Available Items, select Farm, Crop, Farmer, Buyer

Reports, and Dashboards and move them to Selected Items. Click Next.

● From Available

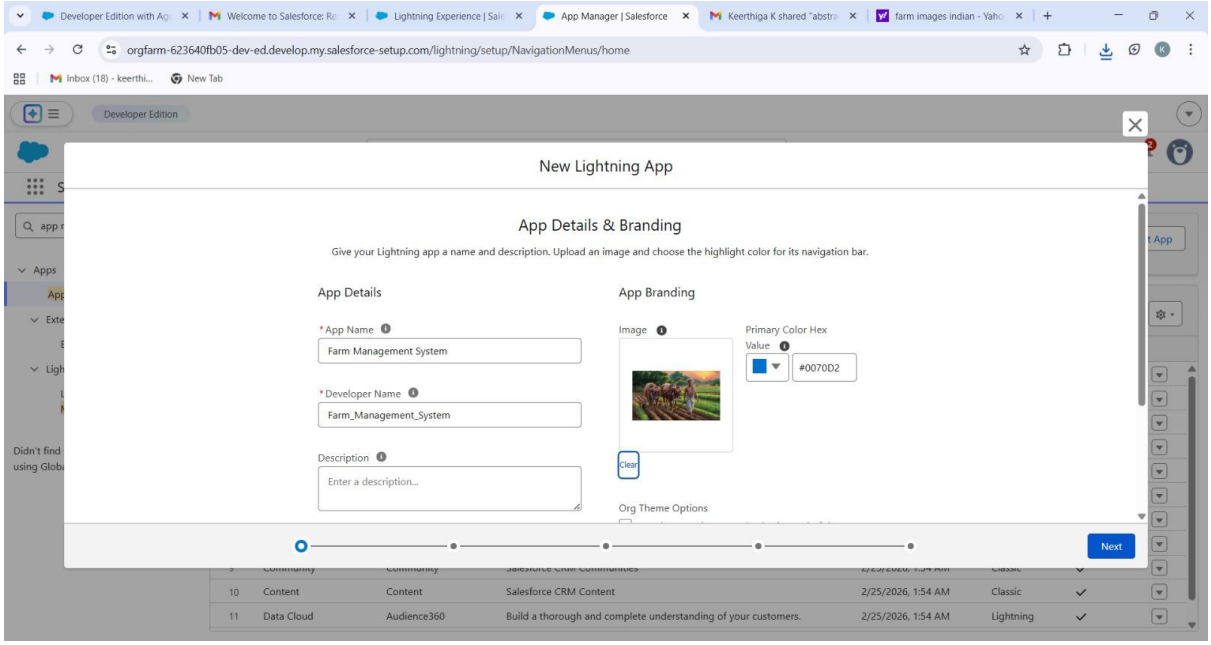
Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.

● To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

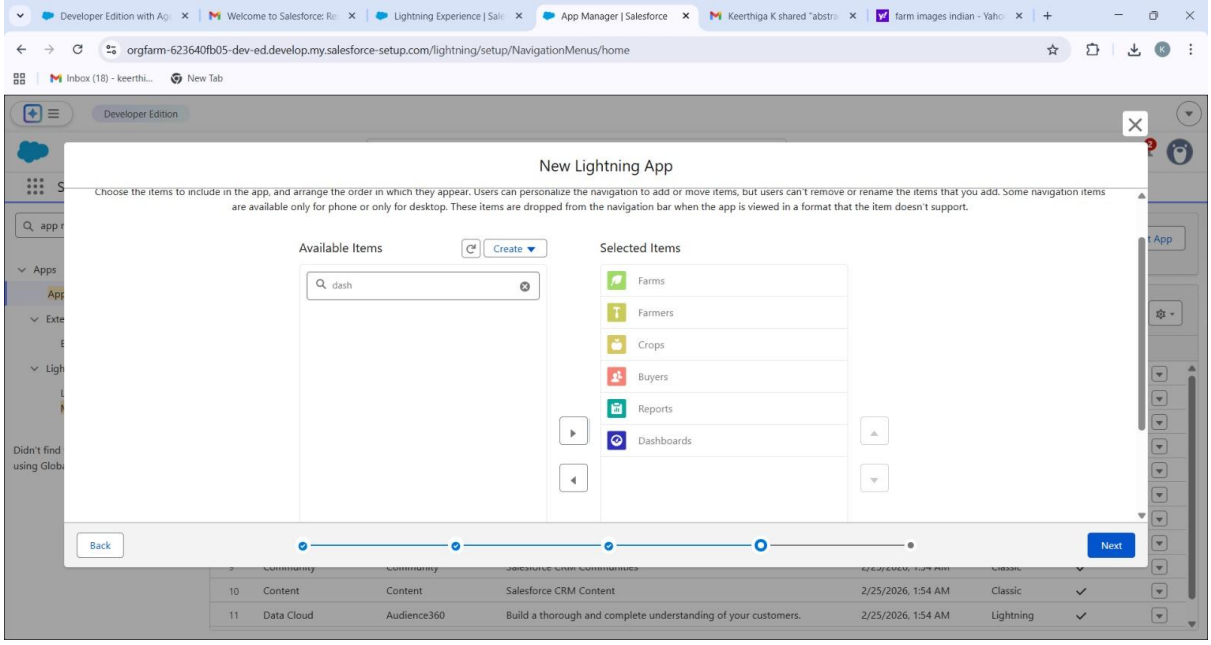
Note:

1. App Launcher-Displays available apps.

2. App Name-Displays the current selected app.



3.Navigation menu-Displays the tabs available inside the app.



MILESTONR-5

Fields and Relationship:

Fields

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

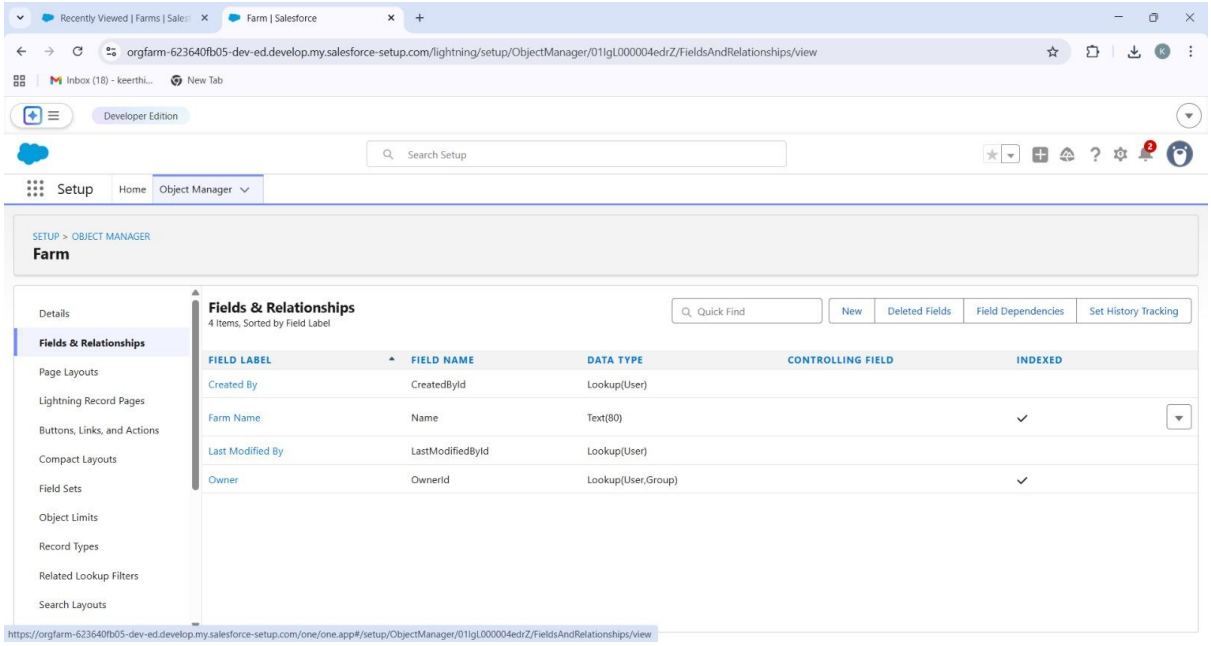
Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.

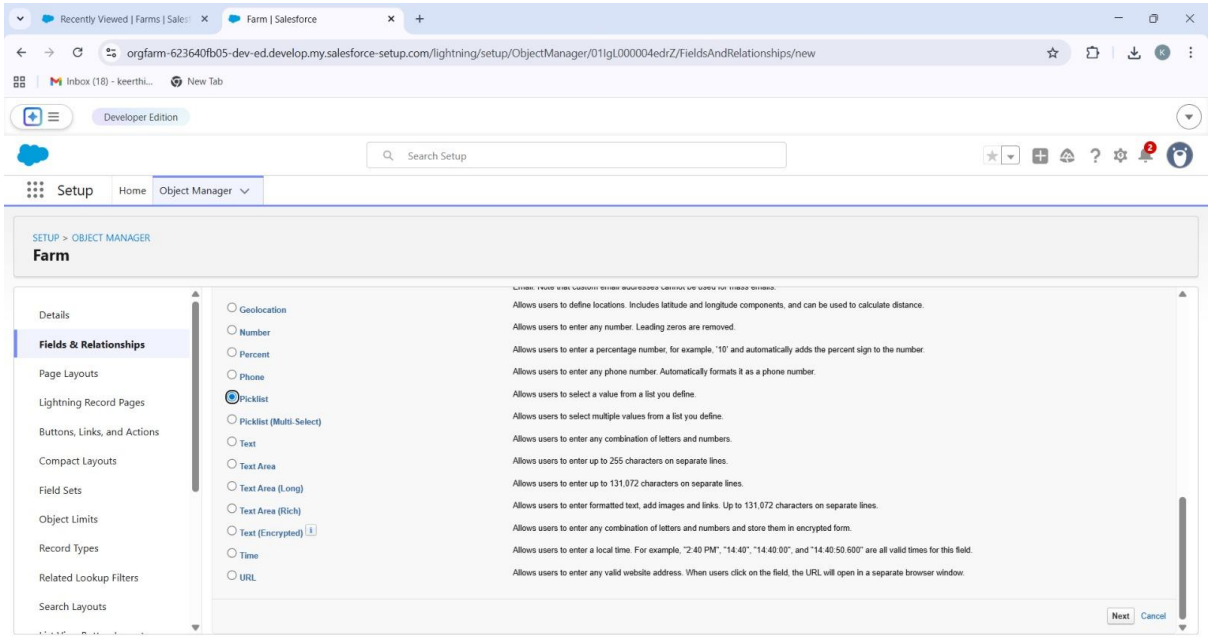
2. Click the Object Manager tab next to Home.

3. Select Farm.

4. Select Fields & Relationships from the left navigation, and click New

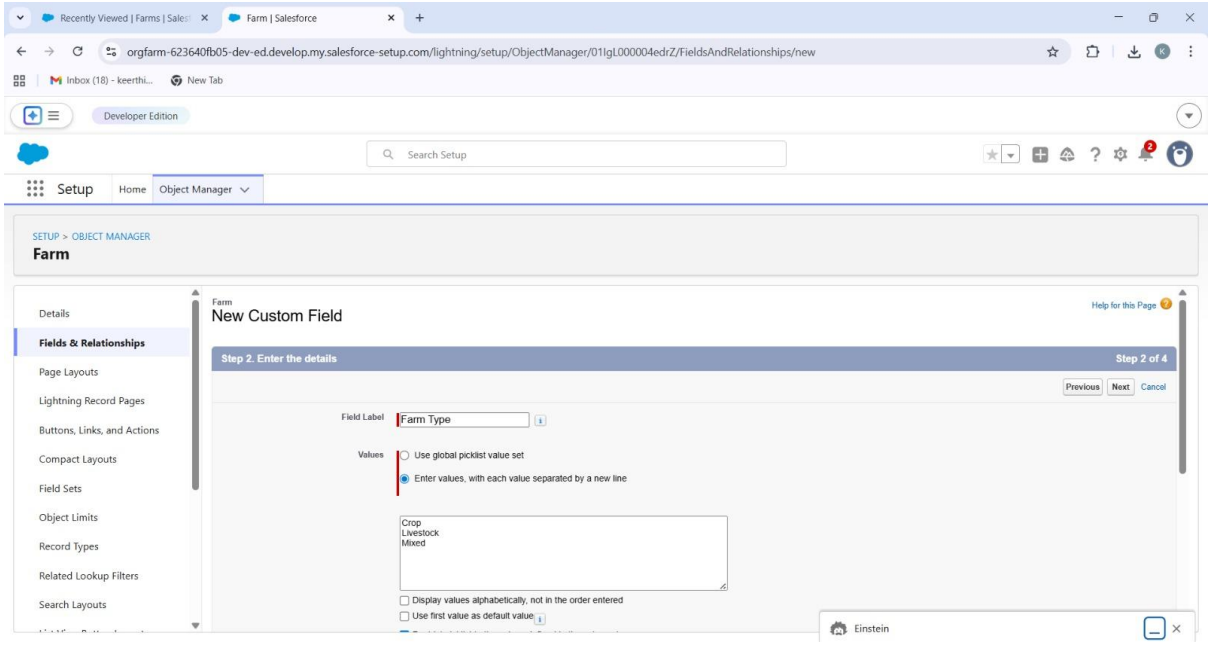


5.Choose Data type as a picklist, Click next



6.Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)

7.Next, Next Click Save



Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

1.Next we Create another Picklist Field, Enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)

2.Again we Create Picklist field, Field Label (Irrigation type), Enter Values (Sprinkler irrigation, Drip irrigation, localized irrigation)

3.Choose data type as a text, enter field label (Location) 4. Choose data type as text, enter field label (size)

5.Create a Master Detail Relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

1.Choose Data type As a text, Enter Field Label name (Crop Name), Next save. 2.Select Data type as date, Enter Field label (Planting Date), then save.

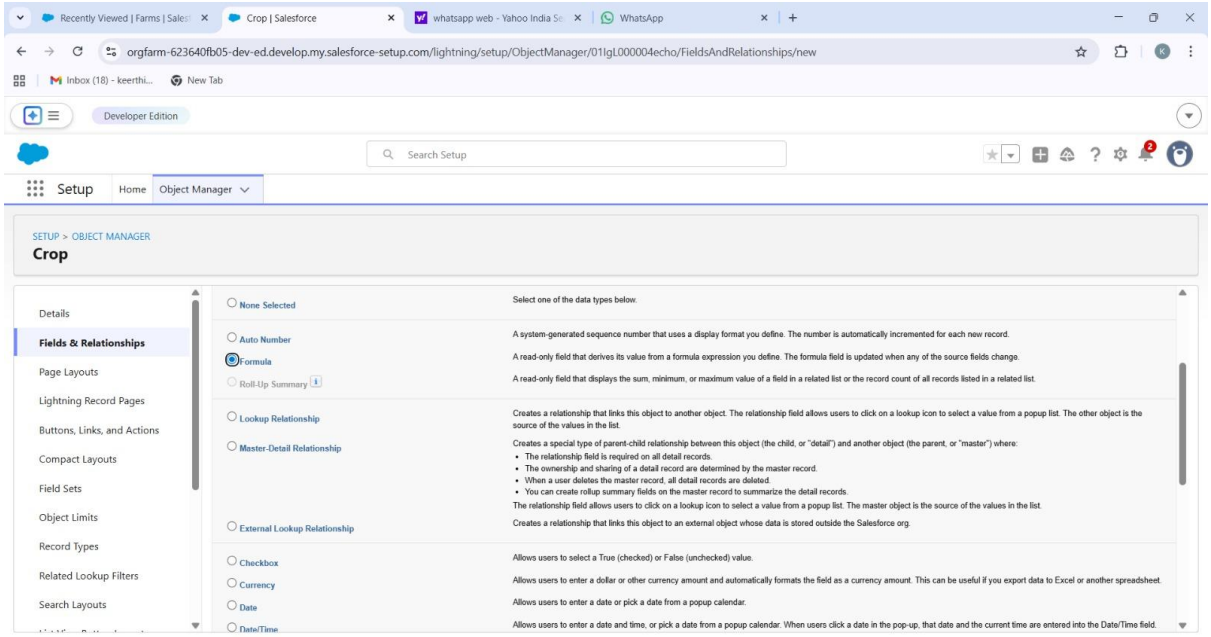
3.Again we select data type as a date, Enter field label (Harvest date).

4.Next We Create Lookup relationship on crop object related to Farm Object.

5.We select a data type as currency, Enter field Label (Actual Yield), Next Save it.

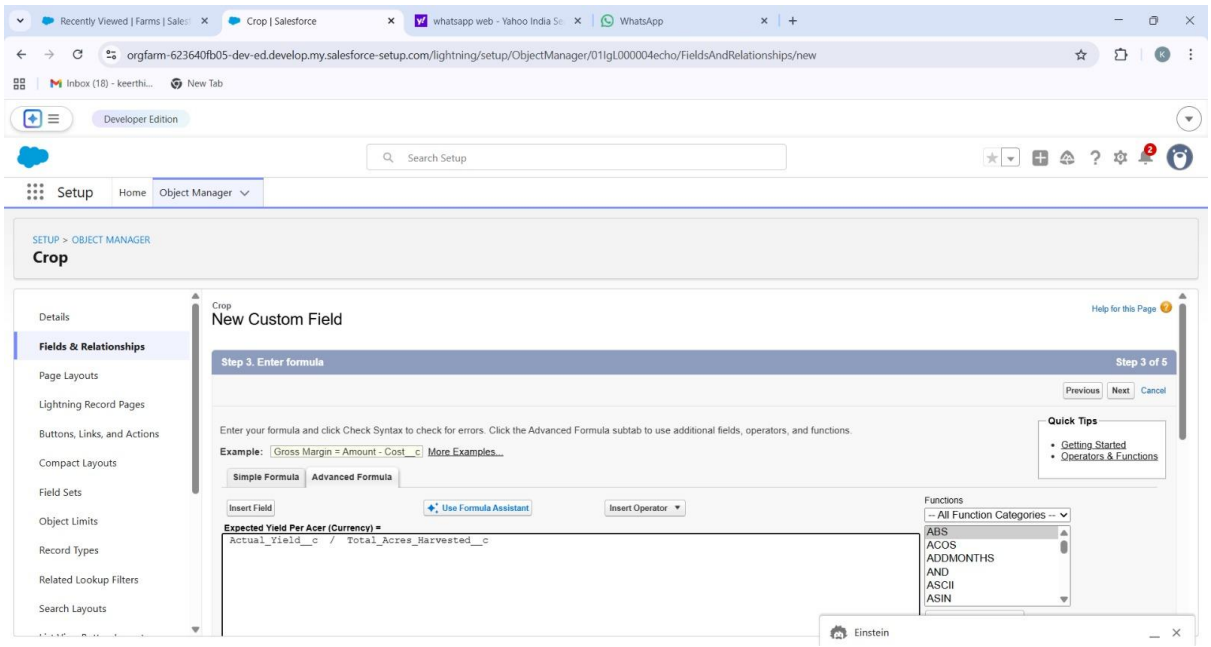
6.Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length (20) and save.

7.Now We create Formula Field For That Select data Type As a Formula, Click Next.



8.Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9.Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity 3:

Creation of fields for the Farmer objects:

1.Select Data type Phone, Enter field label (Contact Number), next Save.

2.Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamil Nadu) , save.

3.Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.

4.Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1.Now create Field Dependency For District field, Open Object Manager, Choose Farmer Object, select Fields& Relationships, On Right side We have Field Dependency option, select that option.

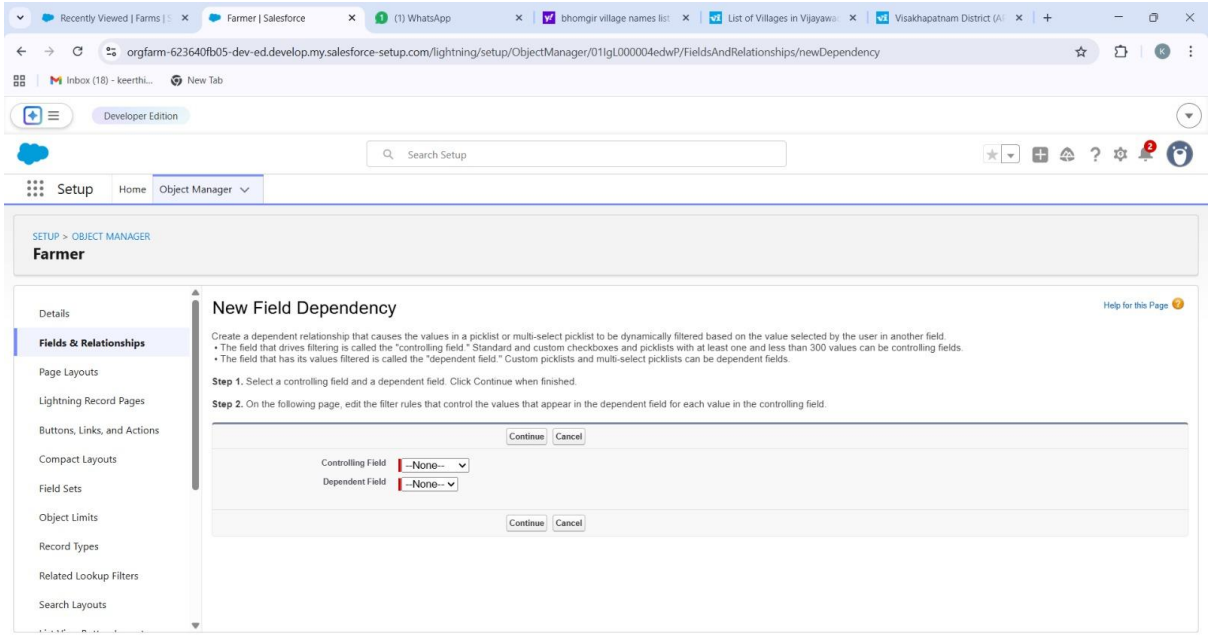
2.Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.

3. Select the appropriate Value in each column by double-clicking them.

4. For Ex. Telangana – Yadadri Bhongir.

5.Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.

6.Click Preview, then test the dependency by selecting a different State and viewing the Associate Values available for particular state.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bhomgir village names list
List of Villages in Vijayawa
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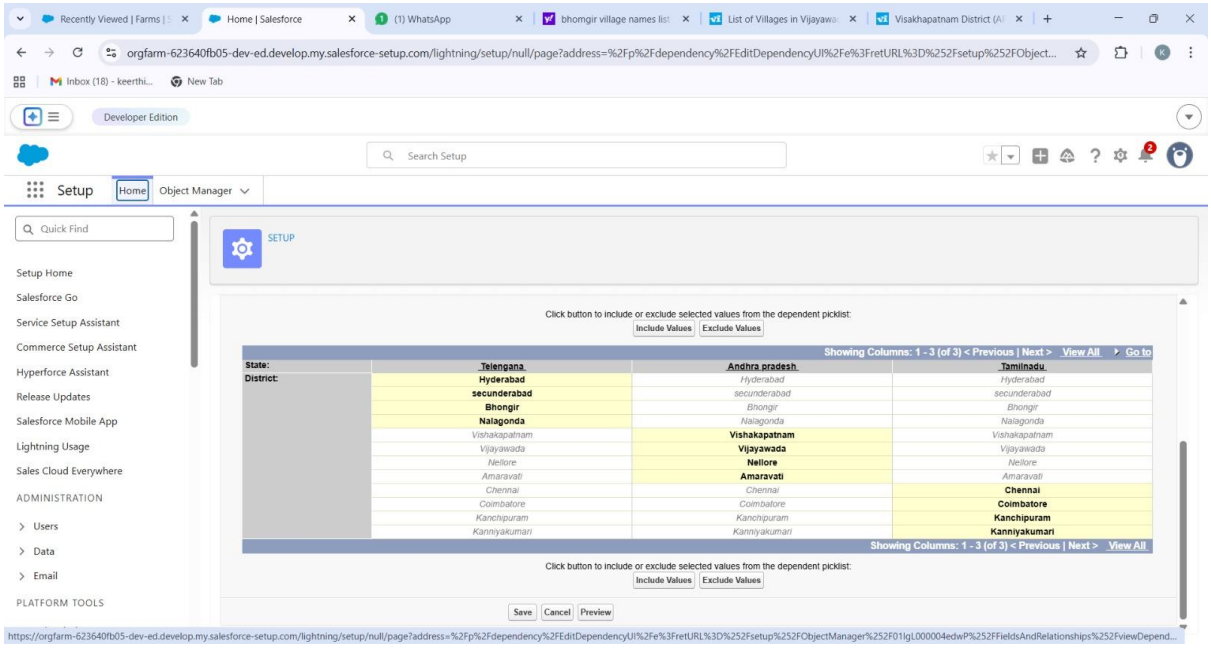
New Tab
Developer Edition
Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Farmer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
New Field Dependency
Help for this Page
Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.
• The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 300 values can be controlling fields.
• The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.
Step 1. Select a controlling field and a dependent field. Click Continue when finished.
Step 2. On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.
Continue
Cancel
Controlling Field
--None--
Dependent Field
--None--
Continue
Cancel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bhomgir village names list
List of Villages in Vijayawa
Visakhapatnam District (Af
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New Tab
Developer Edition
Search Setup
Setup
Home
Object Manager
Quick Find
Setup Home
Salesforce Go
Service Setup Assistant
Commerce Setup Assistant
Hyperforce Assistant
Release Updates
Salesforce Mobile App
Lightning Usage
Sales Cloud Everywhere
ADMINISTRATION
Users
Data
Email
PLATFORM TOOLS
SETUP
Click button to include or exclude selected values from the dependent picklist:
Include Values
Exclude Values
Showing Columns: 1 - 3 (of 3) < Previous | Next > View All Go to
State:
Telengana
Andhra pradesh
Tamilnadu
District:
Hyderabad
secunderabad
Bhongir
Nalagonda
Vishakapatnam
Vijayawada
Nellore
Amaravati
Chennai
Coimbatore
Kanchipuram
Kanniyakumari
Hyderabad
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Bhongir
Nalagonda
Vishakapatnam
Vijayawada
Nellore
Amaravati
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Vishakapatnam
Vijayawada
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Chennai
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Kanniyakumari
Showing Columns: 1 - 3 (of 3) < Previous | Next > View All
Click button to include or exclude selected values from the dependent picklist:
Include Values
Exclude Values
Save
Cancel
Preview
https://orgfarm-623640fb05-dev-ed.develop.my.salesforce-setup.com/lightning/setup/null/page?address=%2Fp%2Fdependency%2FEditDependencyUI%2Fe%3FretURL%3D%252Fsetup%252FObjectManager%252F01IgL000004edwP%252FFieldsAndRelationships%252FviewDepend...

7. Click close to close the preview window, click save.

Activity 4:

Creation of fields for the Buyer objects:

1.Select Data type as a Phone, Enter a Field Label (Contact), Save.

2.Select Data type as an Email, Enter Field Label (Email), Next save.

3.Select Data type Text area, Enter Field Label (Address), Save it.

4.Select Data type Picklist, Field Label (Buyer Type), Enter values

(Wholesale, Retail), Click Save.

MILESTONE-6/ profile

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls "Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

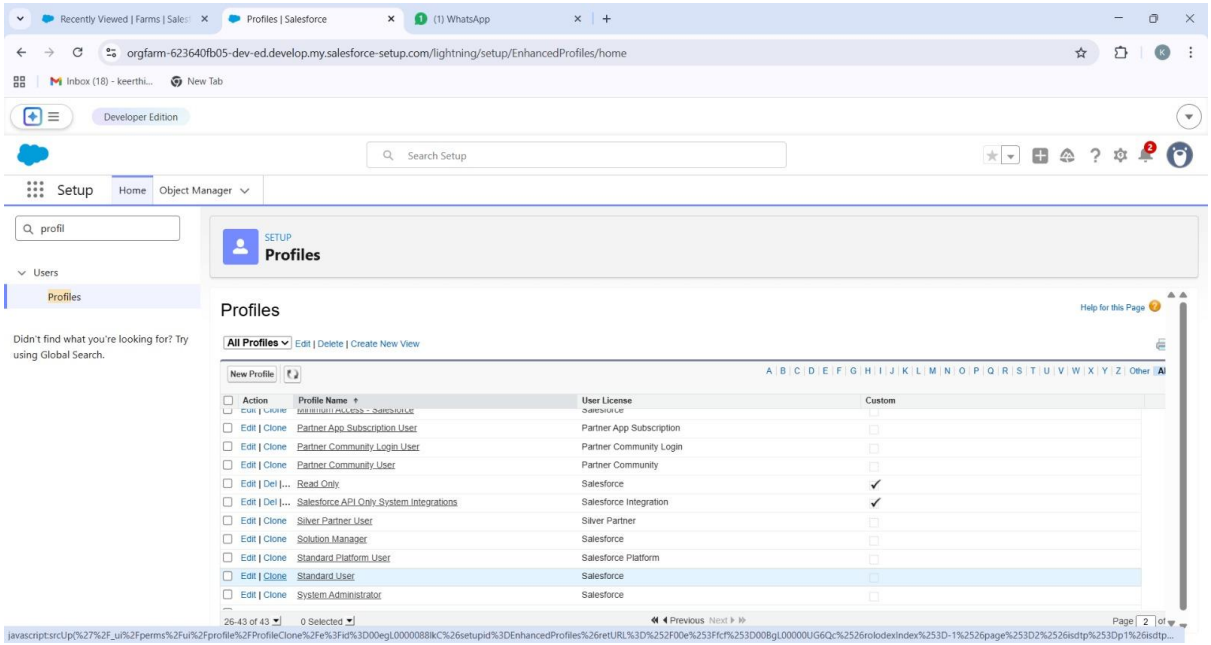
Create a custom profile:

1. From setup, enter profiles in Quick Find box

2. Select profiles (Standard user).

3. Click clone.

4. For Profile, enter Sales Department.


Profiles | Salesforce
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Developer Edition
Search Setup
Setup
Home
Object Manager
profil
Users
Profiles
Didn't find what you're looking for? Try using Global Search.
SETUP
Profiles
Help for this Page
All Profiles
Edit | Delete | Create New View
New Profile
Action
Profile Name
User License
Custom
Partner App Subscription User
Partner Community Login User
Partner Community User
Read Only
Salesforce API Only System Integrations
Silver Partner User
Solution Manager
Standard Platform User
Standard User
System Administrator
Salesforce
Partner Community Login
Partner Community
Salesforce Integration
Silver Partner
Salesforce Platform
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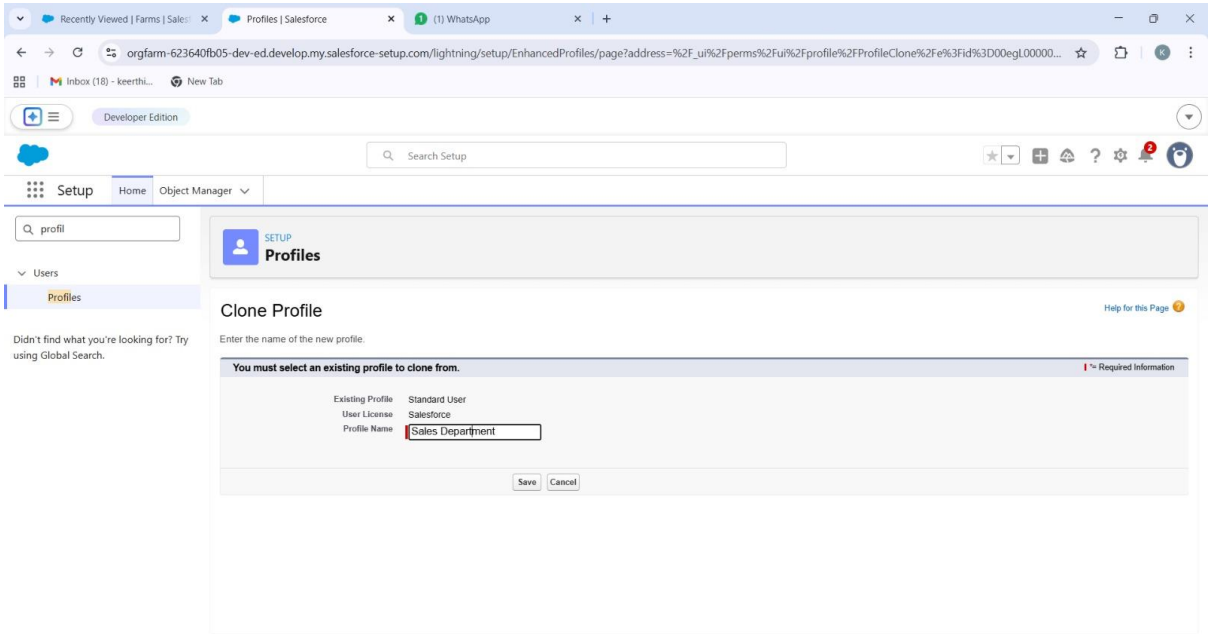
Profiles | Salesforce
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Developer Edition
Search Setup
Setup
Home
Object Manager
profil
Users
Profiles
Didn't find what you're looking for? Try using Global Search.
SETUP
Profiles
Clone Profile
Help for this Page
Enter the name of the new profile.
You must select an existing profile to clone from.
Required Information
Existing Profile
Standard User
User License
Salesforce
Profile Name
Sales Department
Save
Cancel

5.click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “Farm Manager”.

6. From setup, enter profiles in Quick Find box

7. Select profiles (Standard user).

8. Click clone.

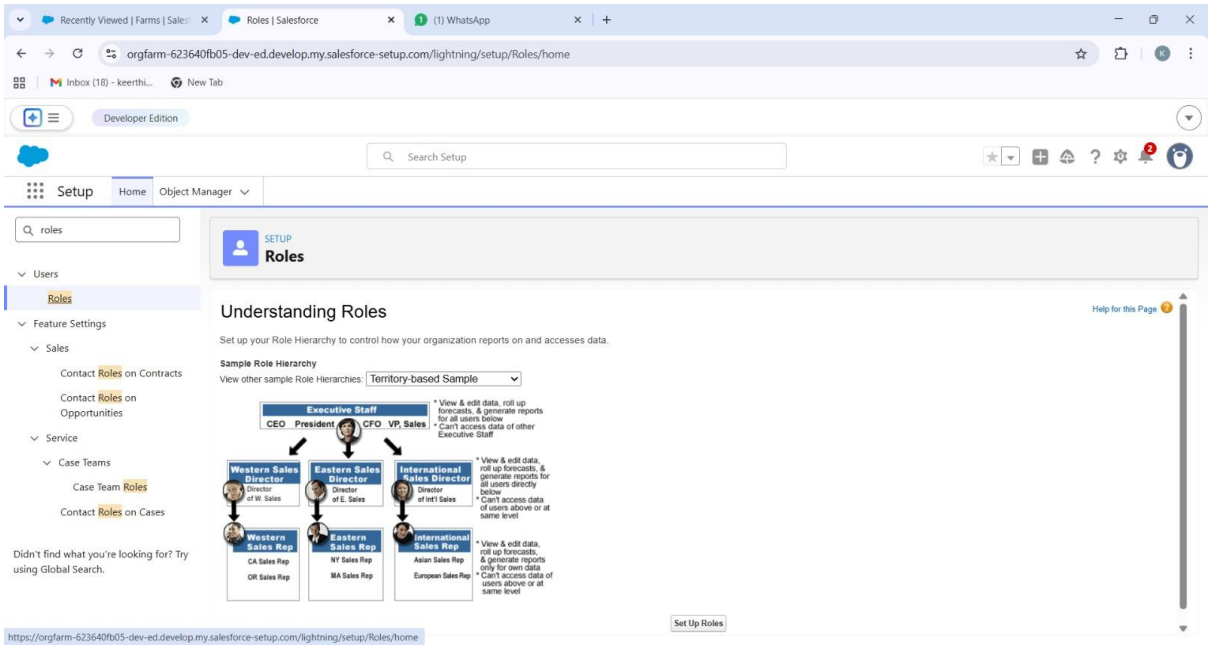
MILESTONE-7/Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

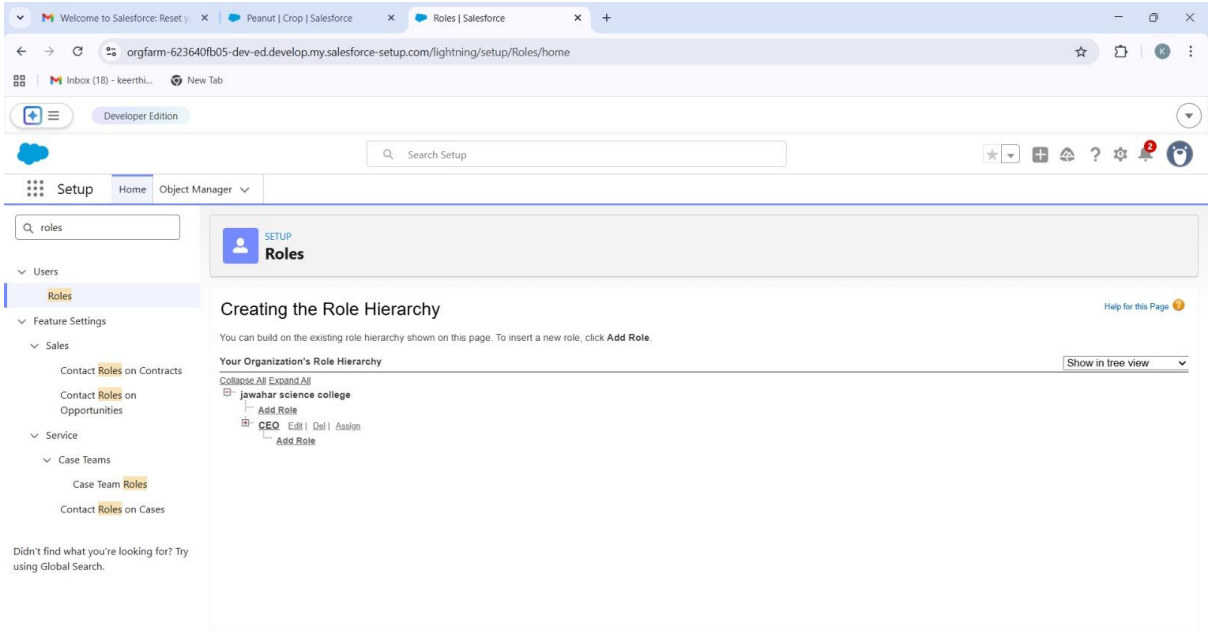
Activity-1

Creation of role

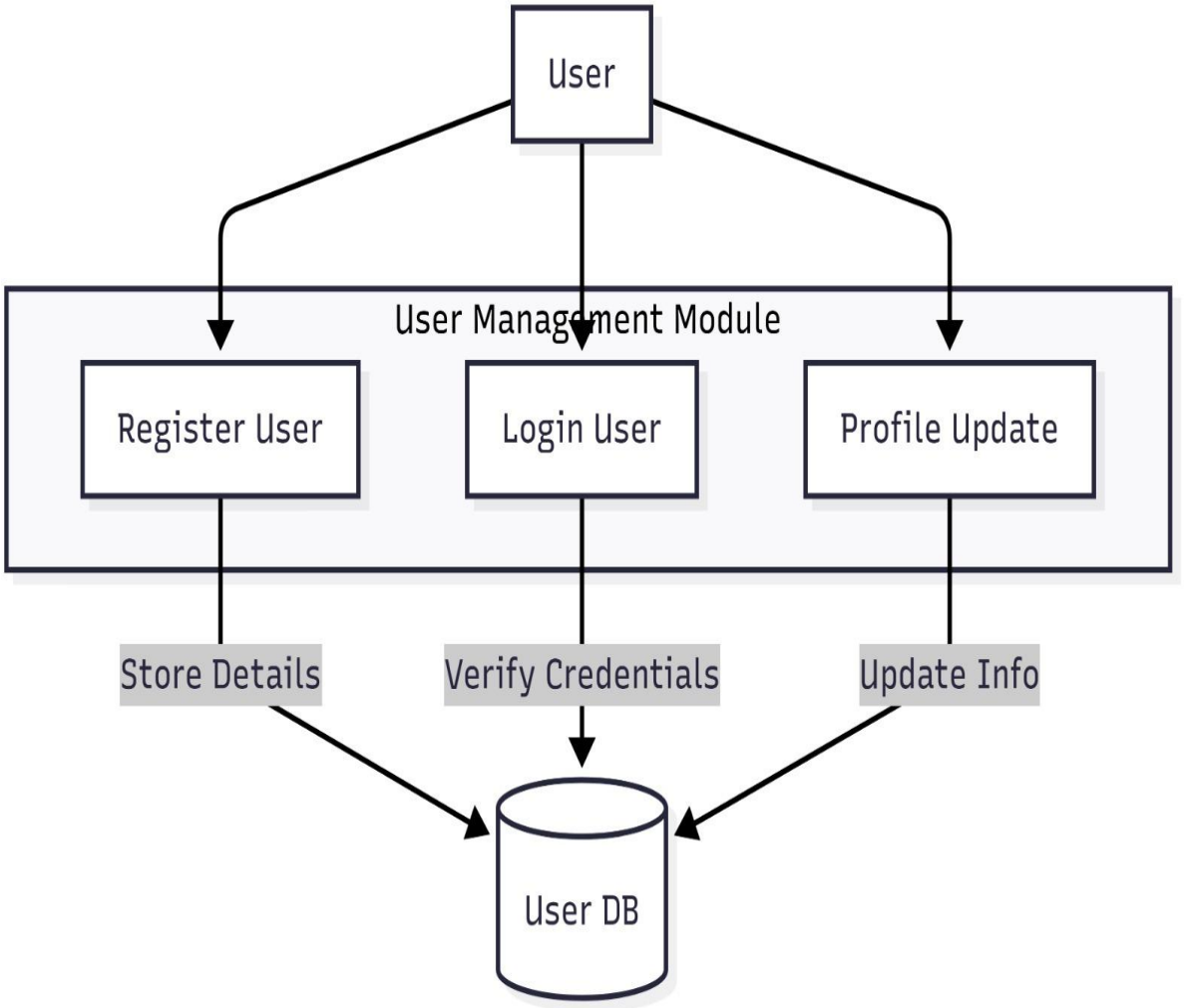
1. From the Quick find box search for the role and click on the roles option
2. Select the setup role option



3.Below the CEO click on add role and enter the label name as a **“Farm Manager”** and role name will be Automatically populated and click on save.


Welcome to Salesforce: Reset y
Peanut | Crop | Salesforce
Roles | Salesforce
orgfarm-623640fb05-dev-ed.develop.my.salesforce-setup.com/lightning/setup/Roles/home
Inbox (18) - keerthi...
New Tab
Developer Edition
Search Setup
Setup
Home
Object Manager
roles
Users
Roles
Feature Settings
Sales
Contact Roles on Contracts
Contact Roles on Opportunities
Service
Case Teams
Case Team Roles
Contact Roles on Cases
Didn't find what you're looking for? Try using Global Search.
SETUP
Roles
Creating the Role Hierarchy
Help for this Page
You can build on the existing role hierarchy shown on this page. To insert a new role, click Add Role.
Your Organization's Role Hierarchy
Show in tree view
Collapse All Expand All
jawahar science college
Add Role
CEO Edit | Del | Assign
Add Role

USERS


User
User Management Module
Register User
Login User
Profile Update
Store Details
Verify Credentials
Update Info
User DB

MILESTONE-8/USER

User

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

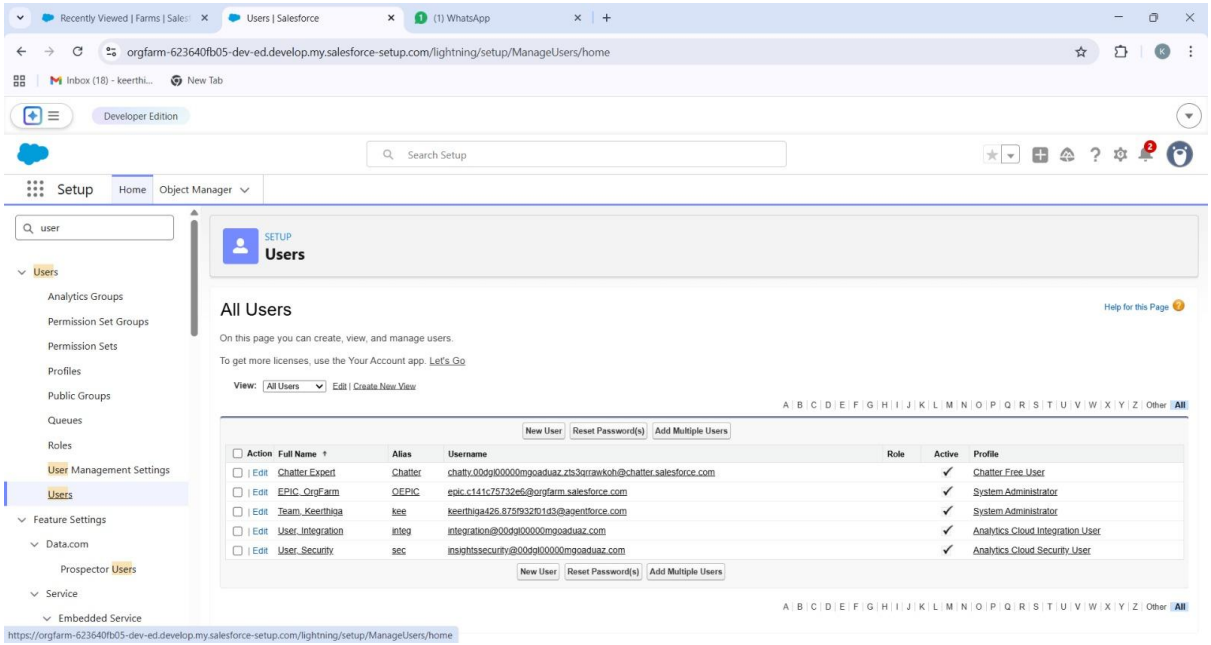
Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and record the user can access.

Activity 1:

To Create a user:

1.From Setup, enter Users in the Quick Find box, then select Users.

2.Click new user



3. Enter First name as Farm and last name as Manager.

4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.

5. Select the role of Farm Manager.

6. Select user License as Standard Platform User.

7. Select profile (Farm Manager).

8. Click save.

Activity 2:

Create a user with a username as “Sales Department”, and assign him the sales Manager profile. Follow the steps from above Activity

MILSTONE-9/Sharing Rules

Sharing Rules

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

1.Goto Sharing Settings, which can be found under the Quick Find section.

2. Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.

3. Add the label of the sharing rule you want to make.

4. Select your rule type based on the criteria.

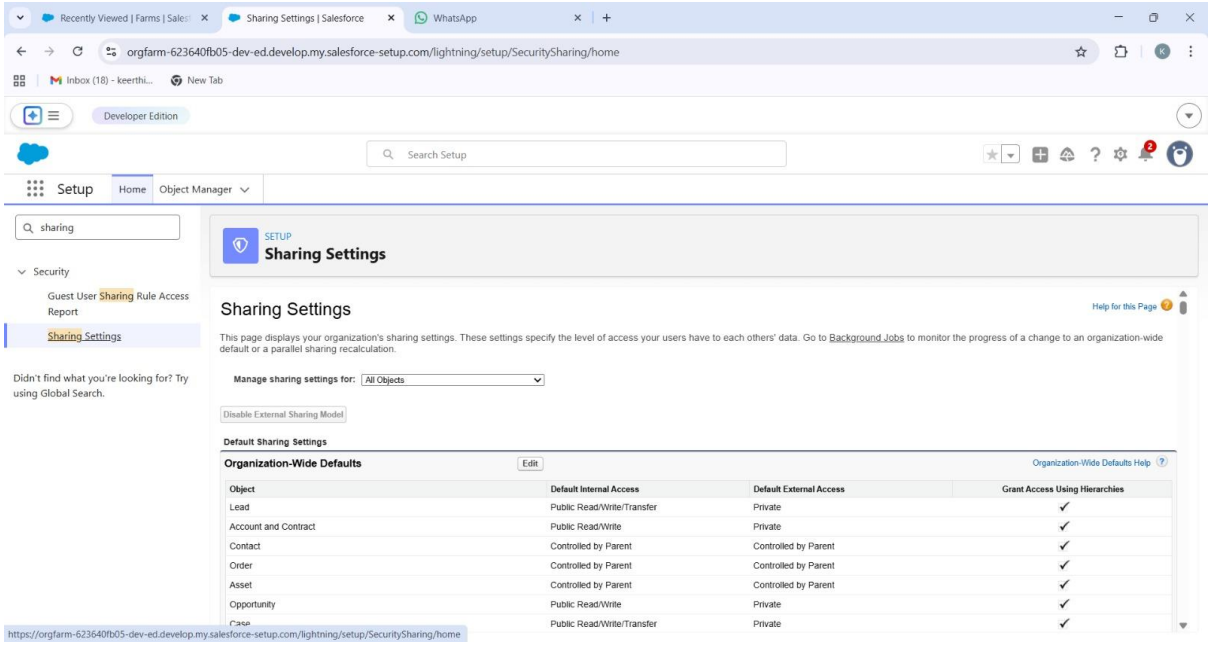
5. Select the field can join immediately check field from the Crop object.

6. Select the Crop as equal and value is True.

7. And in selecting the users to share with the section select roles and in that select Farm Manager.

8. And in the section of select the level of access for the users give the access read/write.

9. And save Rule.



Activity 2:

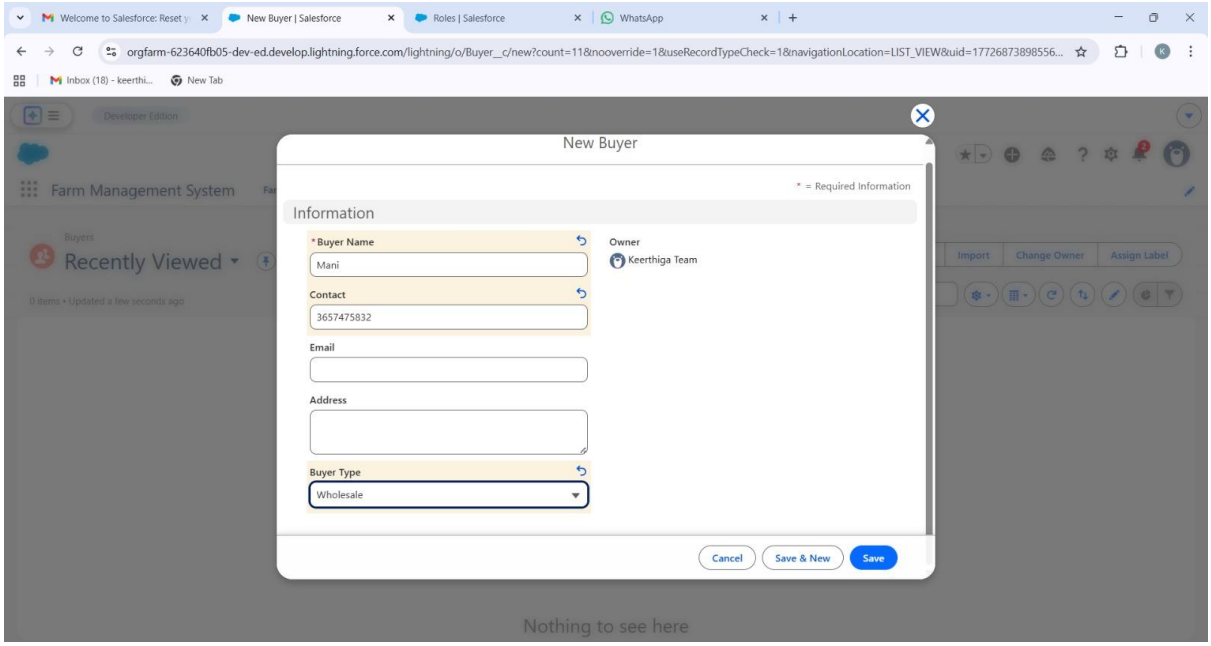
Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record (Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.


Welcome to Salesforce: Reset y
New Buyer | Salesforce
Roles | Salesforce
WhatsApp
orgfarm-623640fb05-dev-ed.develop.lightning.force.com/lightning/o/Buyer__c/new?count=11&nooverride=1&useRecordTypeCheck=1&navigationLocation=LIST_VIEW&uid=17726873898556...
Inbox (18) - keerthi...
New Tab
Developer Edition
Farm Management System
Buyers
Recently Viewed
0 items • Updated a few seconds ago
Import
Change Owner
Assign Label
New Buyer
* = Required Information
Information
* Buyer Name
Mani
Contact
3657475832
Email
Address
Buyer Type
Wholesale
Owner
Keerthiga Team
Cancel
Save & New
Save
Nothing to see here

REPORT

MILESTONE-11/Reports:

Report

Reports in Salesforce are tools used to collect, organize, and display data stored in objects in a structured format. They help users analyse information and understand business activities by presenting data in tables, summaries, or charts.

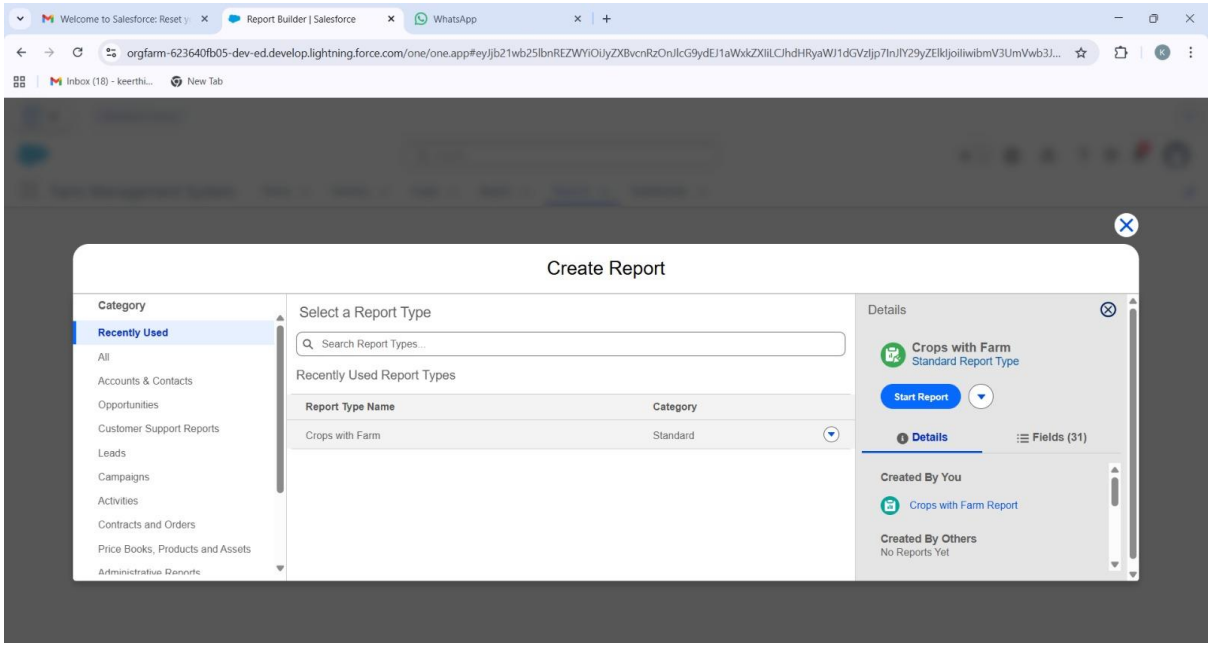
Activity 1:

Create a report:

Create a report that displays rating of the account, and which has type and account name.

1) Click on app launcher search for reports.

2) Click on the new report and select the category has Farmer with Farms.

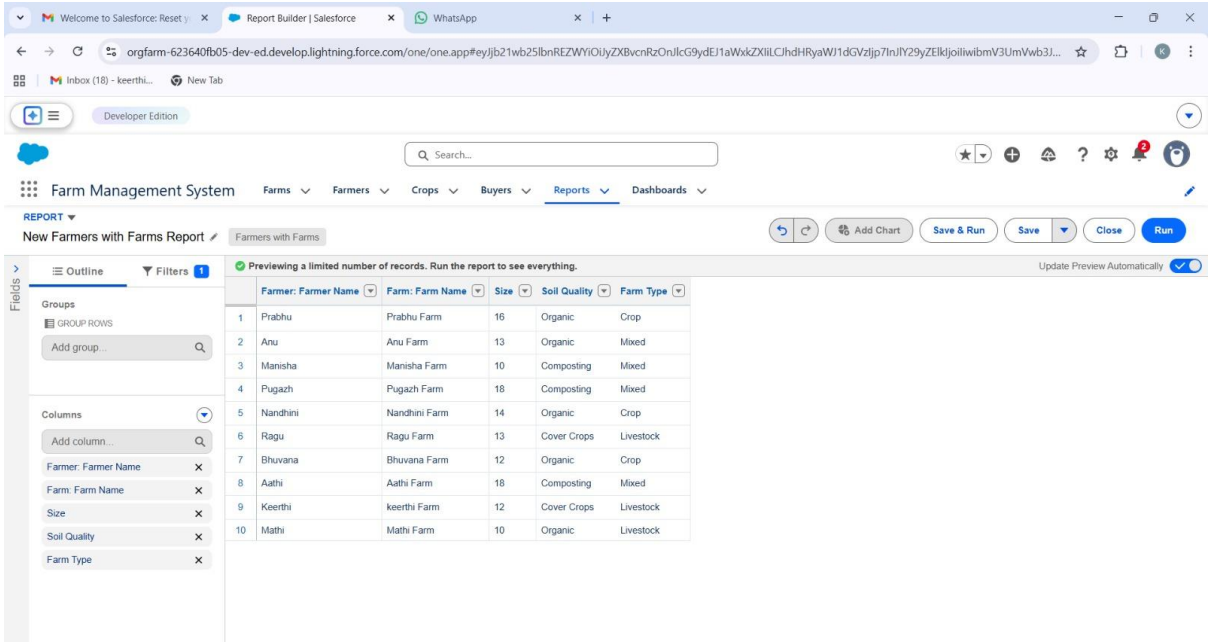


3) In the details section select the option start report.

4) In Outline Click on Column Select Which Field you want to display in the Report.

5) Click Group, Select Which Fields You Want to Group.

6) Save the report by giving label name (Farmer with Farms) and save the folder as a public folder and save the report.



	Farmer: Farmer Name	Farm: Farm Name	Size	Soil Quality	Farm Type
1	Prabhu	Prabhu Farm	16	Organic	Crop
2	Anu	Anu Farm	13	Organic	Mixed
3	Manisha	Manisha Farm	10	Composting	Mixed
4	Pugazh	Pugazh Farm	18	Composting	Mixed
5	Nandhini	Nandhini Farm	14	Organic	Crop
6	Ragu	Ragu Farm	13	Cover Crops	Livestock
7	Bhuvana	Bhuvana Farm	12	Organic	Crop
8	Aathi	Aathi Farm	18	Composting	Mixed
9	Keerthi	keerthi Farm	12	Cover Crops	Livestock
10	Mathi	Mathi Farm	10	Organic	Livestock

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

DASHBOARD

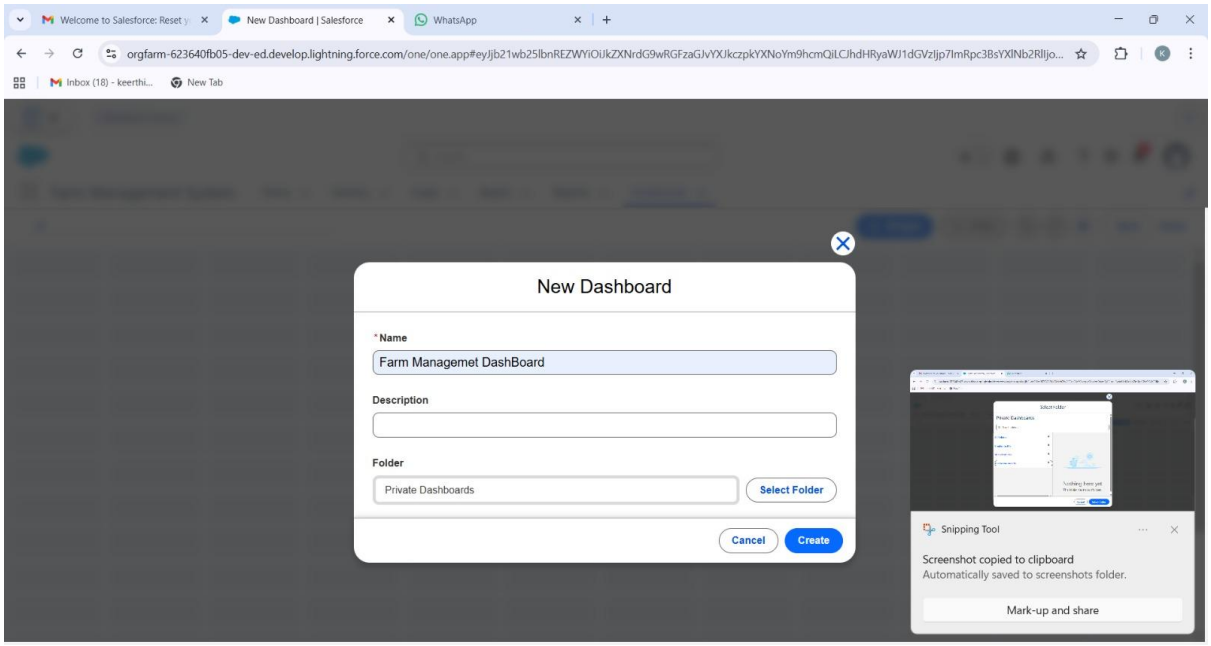
MILESTONE-12/Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



VIEW AND DASHBOARD

MILESTONE-13/ View Report and Dashboard:

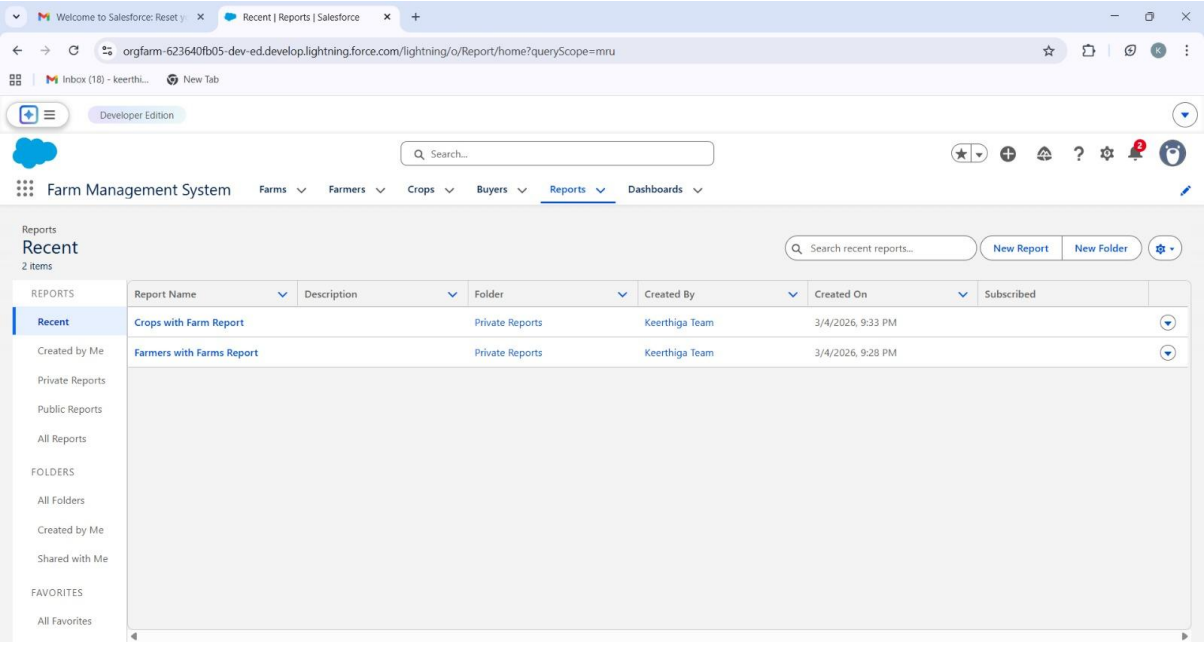
Activity 1: Report

● Click on App Launcher on left side of screen.

● Search Farm Management System & click on it.

● Click on Reports Tab.

● Click on Farmers with Farms & see records



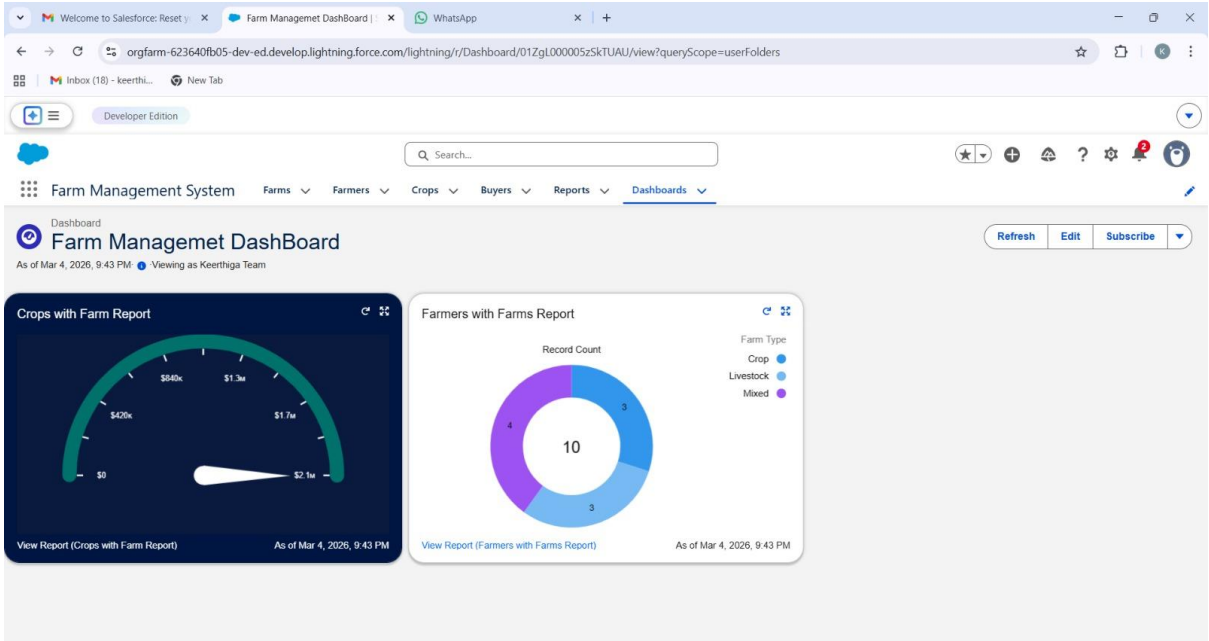
Activity 2: Dashboard

• Click on App Launcher on left side of screen.

• Search Farm Management system & click on it.

• Click on Dashboard Tab.

• Click Farmers With farms & see records.


Welcome to Salesforce: Reset y
Farm Managemet DashBoard |
WhatsApp
orgfarm-623640fb05-dev-ed.develop.lightning.force.com/lightning/r/Dashboard/01ZgL000005zSkTUAU/view?queryScope=userFolders
Inbox (18) - keerthi...
New Tab
Developer Edition
Search...
Farm Management System
Farms
Farmers
Crops
Buyers
Reports
Dashboards
Dashboard
Farm Managemet DashBoard
As of Mar 4, 2026, 9:43 PM · Viewing as Keerthiga Team
Refresh
Edit
Subscribe
Crops with Farm Report
$0
$420k
$840k
$1.3м
$1.7м
$2.1м
View Report (Crops with Farm Report)
As of Mar 4, 2026, 9:43 PM
Farmers with Farms Report
Record Count
Farm Type
Crop
Livestock
Mixed
10
3
3
4
View Report (Farmers with Farms Report)
As of Mar 4, 2026, 9:43 PM

FUNCTIONAL & PERFORMANCE TESTING

During the testing stage, screenshots were collected from the Salesforce interface to confirm that the system features were functioning properly. The following functionalities were verified:

- Creation of records for Farm, Crop, Farmer, and Buyer
- Validation of required input fields
- Verification of relationships and dependencies between objects
- Testing user access through profiles and roles
- Checking sharing rules and permissions
- Generating reports and dashboards successfully
- Performance testing was conducted to ensure the system responds efficiently and remains stable when multiple users operate the system at the same time.

ADVANTAGES & DISADVANTAGES

Advantages

- Centralized storage of agricultural data
- Minimizes manual paperwork
- Higher data accuracy and reliability
- Easy monitoring of farm activities
- Access to real-time reports and analytics

Disadvantages

- Requires stable internet connectivity
- Dependence on Salesforce platform licensing
- Limited functionality in offline mode

CONCLUSION

The **Farm Management System** (FMS) developed using the Salesforce platform provides an efficient and organized solution for managing agricultural data. The system successfully digitizes important farming information such as farms, crops, farmers, and buyers, and stores it in a centralized cloud-based environment.

By implementing Salesforce features like custom objects, relationships, validation rules, reports, and dashboards, the application ensures accurate data management and easy access to information. The system reduces manual paperwork, minimizes errors, and improves the efficiency of farm management activities.

Through the use of reports and dashboards, users can easily monitor farm operations and analyse agricultural data in real time. This helps farmers and managers make better decisions based on accurate and structured information.

Overall, the Farm Management System demonstrates how modern cloud technology can be effectively used to simplify agricultural management processes. The system improves productivity, enhances data visibility, and supports better planning and decision-making in the agricultural sector.

FUTURE SCOPE

Automation Testing Integration

In the future, functional testing can be automated using tools like Selenium to reduce manual effort and improve testing speed.

AI-Based Testing

Artificial Intelligence can be used to automatically detect bugs, predict failures, and optimize testing strategies.

Cloud-Based Performance Testing

Performance testing can be executed in cloud environments using platforms like Amazon Web Services to simulate real-world traffic from different locations.

Continuous Testing in DevOps

Testing can be integrated with DevOps pipelines using tools such as Jenkins to enable continuous testing during software development.

Improved Security Testing Integration

Future testing frameworks may combine functional, performance, and security testing for more reliable applications.

Real-Time Monitoring and Analytics

Advanced monitoring tools can provide real-time analytics to track application performance and detect issues quickly.

Mobile and Cross-Platform Testing

Future applications will require testing across multiple devices, browsers, and operating systems to ensure compatibility.

Scalability Testing Enhancements

Performance testing will focus more on scalability to handle increasing users and data loads.

Integration with Machine Learning

Machine learning can help automatically generate test cases and identify patterns in system failures.

User Experience (UX) Testing Improvements

Future testing may focus more on user experience to ensure applications are not only functional but also user-friendly.

APPENDIX

Source Code: Salesforce configurations and metadata components

Dataset: Records related to Farm, Crop, Farmer, and Buyer

Tools Used: Salesforce Developer Edition

THANK YOU